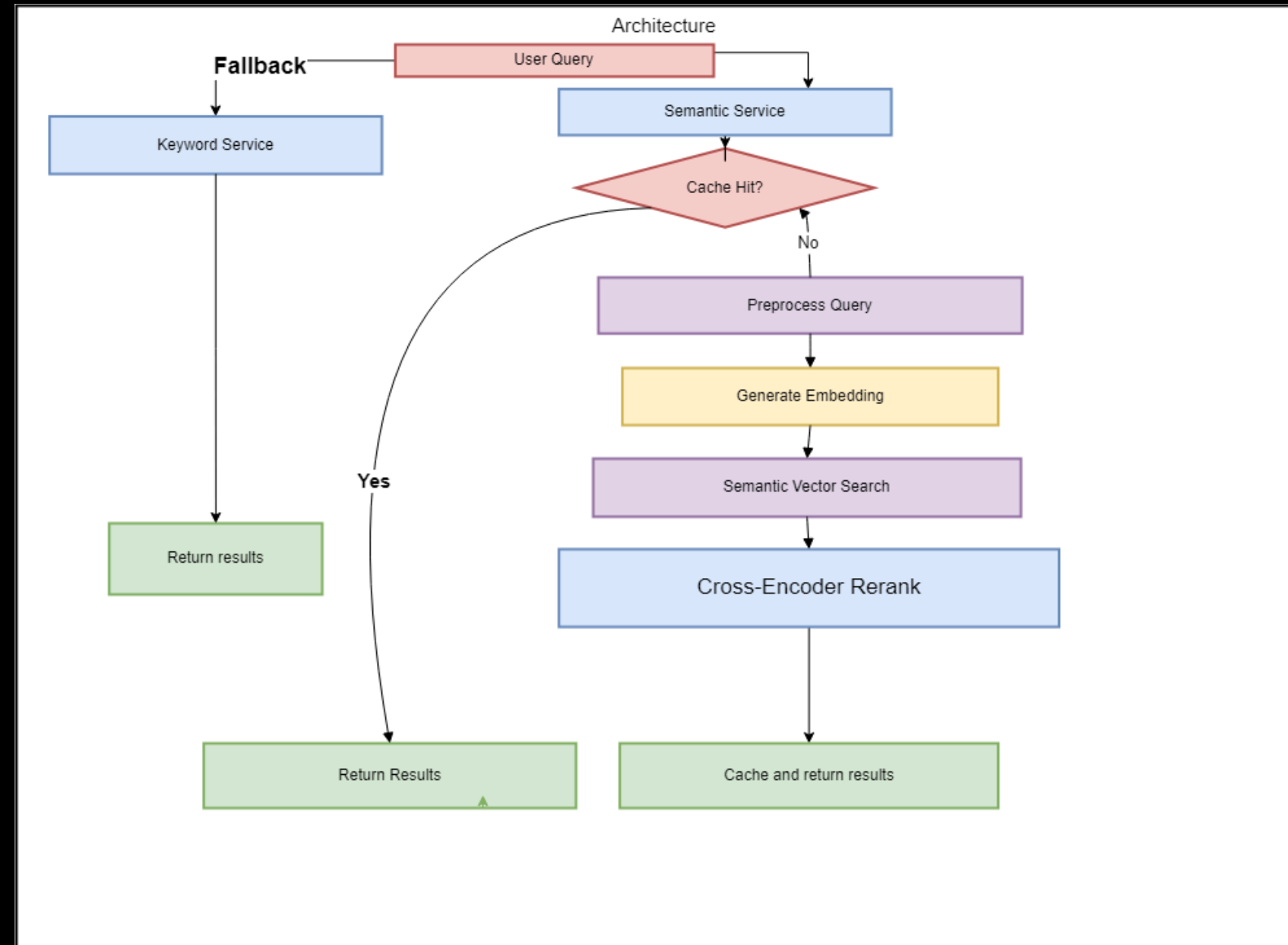


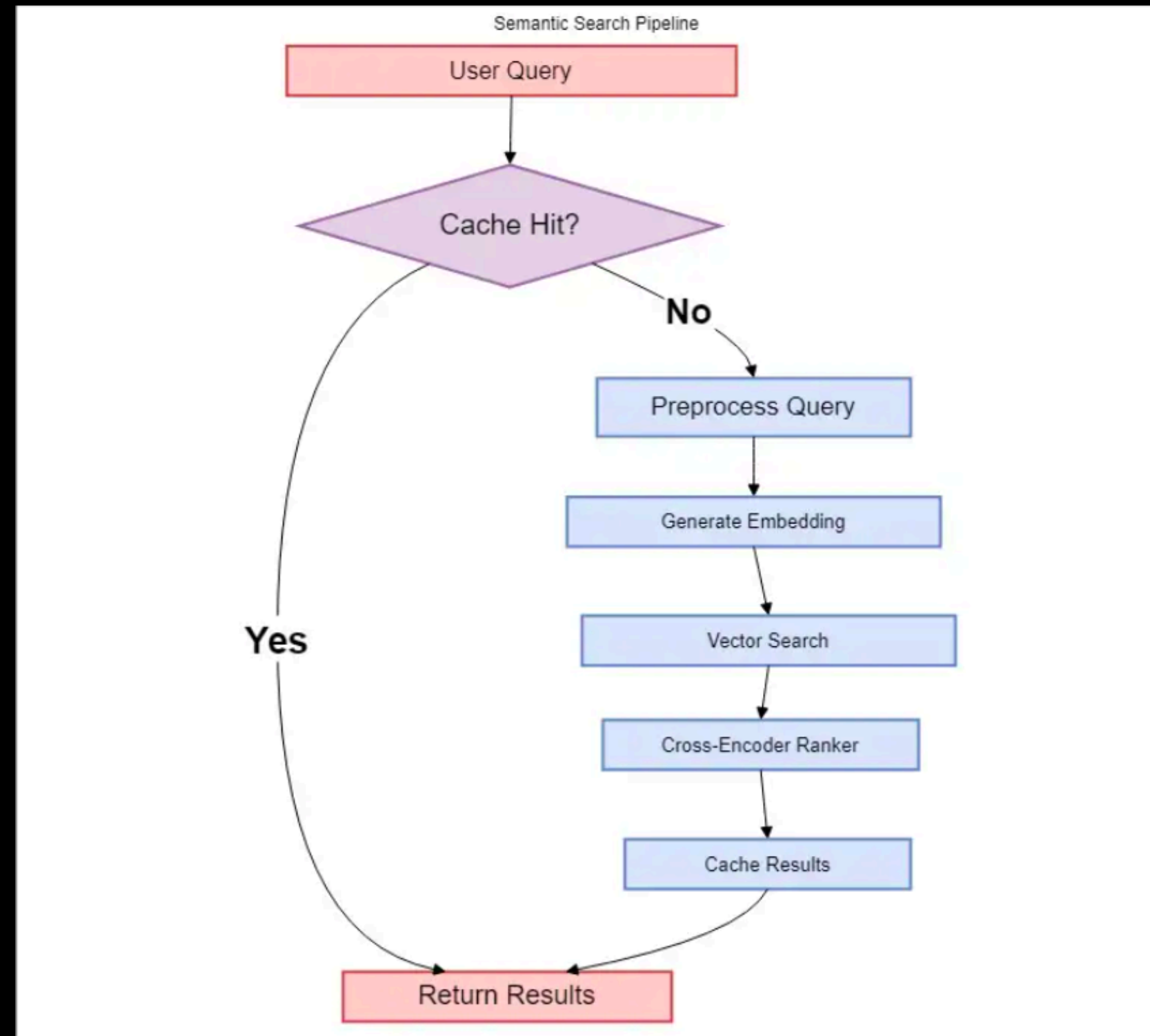
BUET

X-Factor

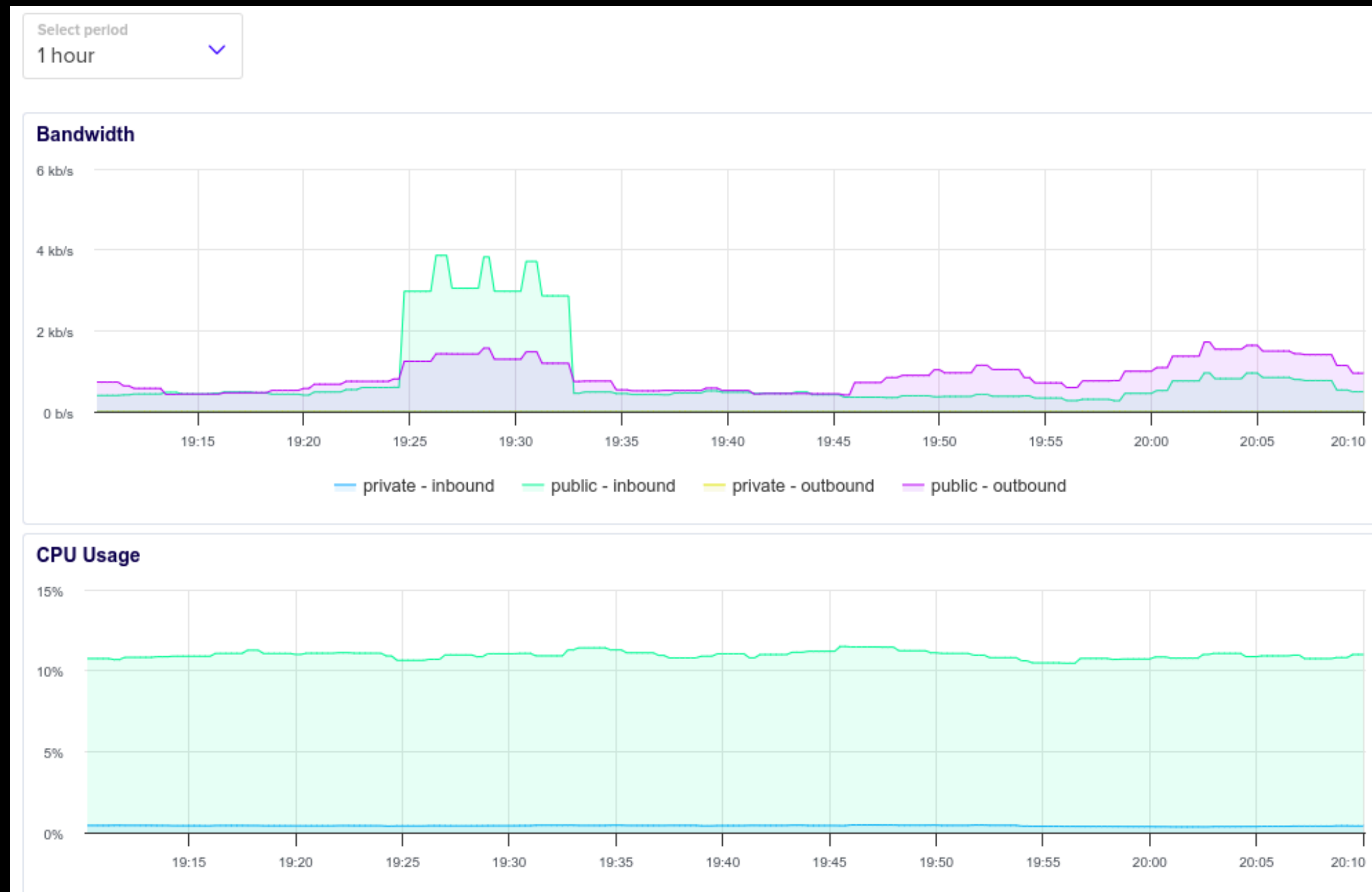
Architecture Diagram



Semantic Search



Airflow Resource Usage



DevOps Overview

CI/CD

Github Actions

- **Continuous Integration on Each PR to Main**
- **Deploy to cloud on each push to Main**

Containerization & Orchestration

Docker

→ Containerized deployment

Kubernetes

→ Container Orchestration

Cloud Infrastructure

DigitalOcean

- **Highly Scalable & Cost efficient**
- **Easy to maintain**

Infrastructure as Code (IaC)

Terraform

- **Infra maintainability**

Other Resources

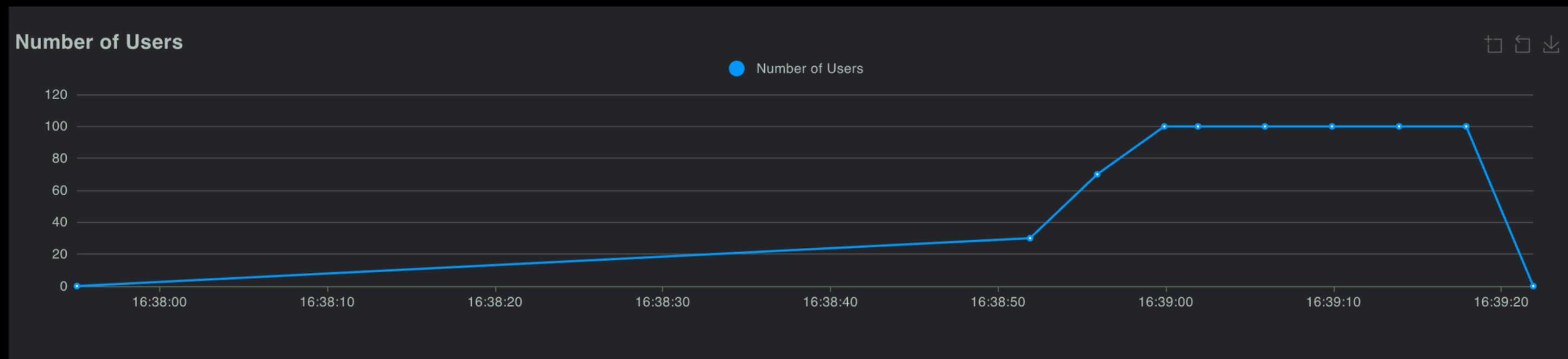
QDrant Cloud

→ Vector DB hosting

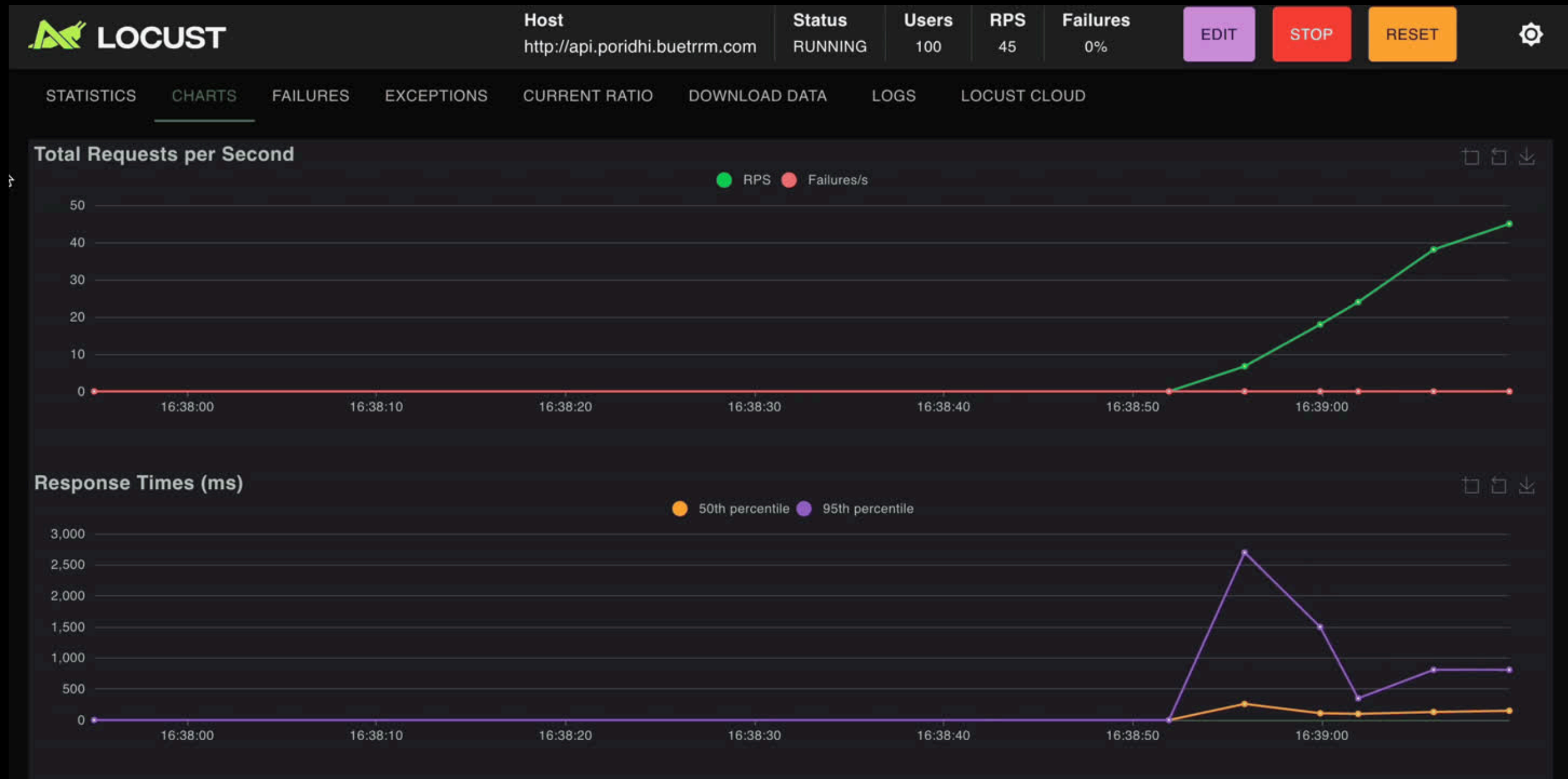
Load Testing

Locust

→ Can easily handle more than 50 concurrent requests.



Load Testing



Load Testing

GET	/keyword-search?query=skirt&page=1	42	0	120	790	950	233.13	85	953	31	1.6	0	ABOUT
GET	/keyword-search?query=socks&page=1	35	0	130	1500	2500	322.59	72	2498	2642	1.2	0	
GET	/keyword-search?query=suit&page=1	36	0	110	790	870	235.86	74	875	1959	1	0	
GET	/keyword-search?query=sweater&page=1	33	0	110	780	910	210.92	73	905	2471	1.8	0	
GET	/keyword-search?query=sweatpants&page=1	44	0	86	810	910	192.33	70	907	576	2.1	0	
GET	/keyword-search?query=swimsuit&page=1	34	0	150	1500	2700	300.69	82	2720	31	1.7	0	
GET	/keyword-search?query=t-shirt&page=1	44	0	120	310	870	178.24	83	870	3748	2	0	
GET	/keyword-search?query=tank%20top&page=1	36	0	100	810	2800	261.16	73	2784	173	1.4	0	
GET	/keyword-search?query=underwear&page=1	39	0	110	800	1500	210.38	75	1479	4459	1.3	0	
GET	/keyword-search?query=vest&page=1	56	0	95	1500	2800	244.65	76	2783	5750	2	0	
Aggregated		1201	0	110	770	1500	221.12	70	3213	1834.6	46.7	0	

```
^Cteflon@pop-os:~/Buet/3-1/CSE310-Compiler$ kubectl get hpa keyword-search-hpa -w
NAME                                REFERENCE                                TARGETS              MINPODS   MAXPODS   REPLICAS   AGE
keyword-search-hpa                 Deployment/keyword-search-service         cpu: 70%/80%         1          4          2           3d14h
keyword-search-hpa                 Deployment/keyword-search-service         cpu: 77%/80%         1          4          2           3d14h
keyword-search-hpa                 Deployment/keyword-search-service         cpu: 152%/80%        1          4          2           3d14h
keyword-search-hpa                 Deployment/keyword-search-service         cpu: 16%/80%         1          4          4           3d14h
keyword-search-hpa                 Deployment/keyword-search-service         cpu: 44%/80%         1          4          4           3d14h
keyword-search-hpa                 Deployment/keyword-search-service         cpu: 5%/80%          1          4          4           3d14h
keyword-search-hpa                 Deployment/keyword-search-service         cpu: 8%/80%          1          4          4           3d14h
keyword-search-hpa                 Deployment/keyword-search-service         cpu: 39%/80%         1          4          4           3d14h
keyword-search-hpa                 Deployment/keyword-search-service         cpu: 64%/80%         1          4          4           3d14h
keyword-search-hpa                 Deployment/keyword-search-service         cpu: 20%/80%         1          4          4           3d14h
keyword-search-hpa                 Deployment/keyword-search-service         cpu: 10%/80%         1          4          4           3d14h
keyword-search-hpa                 Deployment/keyword-search-service         cpu: 5%/80%          1          4          4           3d14h
```

Cloud Cost

**-> With 60-70% of threshold usage -
60 RPS × 60 × 60 × 24 × 30 = ~155 million requests/month**

**→ Assuming 100 Requests/user in a month -
1.55 million users/month**

**→ Infra Cost -
Around 200\$ per month**

Observability

→ **Prometheus**

→ **Grafana**

→ **Promtail**

→ **Loki**

→ **OpenTelemetry**

→ **Jaeger**

Thank You!